Data-Driven Safety Zone Construction from Cardiopulmonary Exercise Testing in Elderly Hypertensive Patients:

R/T/I Three-Axis Framework and Threshold Calibration Using a 2×2 HTN×EIH Design (N=3,206)

TODO: Author names[1]

[1]TODO: Institution

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Abstract

Objective: Cardiopulmonary exercise testing (CPET) is the gold standard for exercise safety assessment, yet data-driven safety zone quantification for Chinese elderly hypertensive patients has not been established. This study aimed to construct a data-driven exercise safety zone based on the R/T/I (Reserve/Threshold/Instability) three-axis framework, develop an improved reference equation adapted to this population, and quantify systematic deviations between data-driven thresholds and published reference values.

Methods: Retrospective single-centre study (N=3,206, ≥60 years) using a hypertension history (HTN) × exercise-induced hypoxaemia (EIH) 2×2 design with four cohorts. The R/T/I axes (cardiopulmonary reserve, ventilatory threshold stability, haemodynamic instability) were combined with equal weighting into a continuous S_{lab} score; safety zone thresholds were determined by reference population percentiles (P75/P90). An improved reference equation incorporated BMI in addition to traditional age/sex variables, validated by 5-fold cross-validation. Bootstrap resampling (1000 iterations) estimated 95% confidence intervals for thresholds, which were systematically compared against Weber-Janicki classification criteria.

Results: The improved reference equation raised VO_2peak R^2 from 0.298 to 0.392 (ΔR^2=+0.094); existing external reference equations yielded $R^2 < 0$ in this population. Bootstrap CI widths for both thresholds were <1.0 (Green/Yellow: 0.65; Yellow/Red: 0.94), indicating highly stable thresholds. The Yellow/Red threshold deviated from Weber-Janicki criteria by +21% (VO_2peak%pred 71% vs 50%), while

the Green/Yellow threshold deviated by only +5%. Reclassification rate was 39.5%, including 188 Red↔Green crossover cases.

Conclusions: This study presents the first data-driven R/T/I three-axis CPET safety zone framework, with Bootstrap-validated robust thresholds (CI width <1.0 for both boundaries). The published Yellow/Red threshold shows a +21% systematic deviation in this elderly hypertensive population, necessitating population-specific calibration. The BMI-enhanced reference equation substantially outperforms existing external equations (ΔR^2=+0.094). This framework provides a reproducible, data-driven paradigm for personalised CPET safety stratification in Chinese elderly hypertensive patients.

Keywords: cardiopulmonary exercise testing; data-driven safety zone; hypertension; exercise-induced hypoxaemia; R/T/I three-axis framework; reference equation; threshold calibration

1 Introduction

Against a backdrop of global ageing, cardiovascular disease (CVD) has become the leading cause of disability and death among older adults [1]. Exercise rehabilitation is well established to improve cardiopulmonary function and long-term prognosis in CVD patients [2, 3]; however, the risk of exercise-induced adverse events—particularly exercise-induced hypoxaemia (EIH) and hypertensive stress responses—substantially constrains the precise implementation of exercise prescriptions, underscoring the need for objective tools to define the exercise safety boundary in elderly cardiovascular patients [4, 5].

Cardiopulmonary exercise testing provides comprehensive cardiopulmonary assessment and represents the gold standard for exercise safety prescription [6, 7]. Existing safety stratification systems, notably the Weber-Janicki functional classification [8], were derived from Western healthy populations or specific disease cohorts and have not been empirically validated in Chinese elderly hypertensive patients through large-scale studies. The Chinese expert consensus on CPET [9] recommends standardised CPET protocols, yet quantitative safety zone frameworks with locally calibrated thresholds remain absent for this population.

Three methodological gaps underlie this limitation. (a) The interaction between HTN history and EIH has not been systematically quantified in large samples, leaving unclear whether these factors operate through independent multi-axis mechanisms that would support a multi-dimensional framework design. (b) Existing reference equations—restricted to age and sex [10–12]—exhibit limited precision in elderly hypertensive patients with pronounced differences in body composition, as BMI has not been incorporated. (c) A data-driven CPET safety zone threshold calibration methodology has not been estab-

lished; a systematic framework to integrate multidimensional CPET information into robust, quantitative stratification is lacking [13].

This study was designed to address these gaps using CPET data from 3,206 elderly hypertensive patients at a single centre. We propose the R/T/I three-axis data-driven safety zone framework, integrating cardiopulmonary reserve (Reserve, R-axis), ventilatory threshold stability (Threshold, T-axis), and haemodynamic instability (Instability, I-axis) into a continuous composite score. We also develop a BMI-enhanced reference equation, formally validate threshold robustness using Bootstrap, and quantify the clinical necessity of population-specific calibration through systematic comparison with Weber-Janicki reference criteria.

2 Methods

2.1 Study Population and Design

This retrospective single-centre cohort study enrolled elderly patients (≥60 years) who underwent CPET between 2015 and 2023. Of 3,232 initial records, 3,206 were included after data quality control (26 excluded, 0.8%). Study design used a 2×2 stratification of HTN history (present/absent) × EIH response (present/absent), yielding four cohorts (Table 1). The study was approved by the institutional ethics committee (approval number: TODO), retrospective design with waived informed consent, and all data were de-identified.

Table 1: HTN×EIH 2×2 cohort design

Cohort	Code	N (post-QC)	Description
Control / non-HTN	CTRL	1,858	No HTN history, no EIH
HTN history, no EIH	HTN-NoEIH	738	HTN history, no EIH during testing
HTN history with EIH	HTN-EIH	277	HTN history, EIH during testing
EIH only	EIH-Only	333	No HTN history, EIH during testing
Total		3,206	(raw: 3,232)

2.2 CPET Protocol and Variable Definitions

CPET was performed using incremental-load cycle ergometer or treadmill protocols with standard gas metabolic analysis. Key measures included: peak oxygen uptake (VO_2peak, mL/kg/min), peak heart rate (HR_{peak}, bpm), oxygen pulse (O_2 pulse, mL/beat), VO_2 at the first ventilatory threshold (VT1-VO_2), V_E/V_{CO_2} slope, heart rate recovery (HRR, 1 minute), oxygen uptake efficiency slope (OUES), and peak METs. EIH was defined as a

$\geq$4% decrease in SpO_2 from rest or an absolute value <90% during exercise (identified by group-coding field).

2.3 Data Quality Control

The quality control pipeline comprised five layers: physiological range compliance (with clip corrections for 35 outliers), cross-field logical consistency, missingness threshold inspection (>30% flagging), protocol consistency (cycle/treadmill standardisation), and EIH-group-specific diagnostics. Twenty-six records (0.8%) were excluded, yielding 3,206 for analysis.

2.4 Improved Reference Equations

Traditional reference equations rely solely on age and sex [10, 11]. Using the reference subset (CTRL group, n=962), we compared four formula complexities (adding BMI as predictor) and selected the optimal formulation based on 5-fold cross-validation R^2 and AIC. The improved reference equation (v2) for VO_2peak is:

$$\widehat{\mathrm{VO_2peak}} = \alpha + \beta_1 \cdot \mathrm{age} + \beta_2 \cdot \mathbf{1}_M + \beta_3 \cdot (\mathrm{age} \times \mathbf{1}_M) + \beta_4 \cdot \mathrm{BMI} \tag{1}$$

Applicability of external reference equations (Wasserman & Hansen 1999, Koch et al. 2009 [12]) was evaluated with R^2, RMSE, and systematic bias to formally test the necessity of population-specific calibration.

2.5 R/T/I Three-Axis Safety Zone Framework

2.5.1 Axis Score Definitions

The safety zone framework comprises three independently scored axes:

R-axis (Reserve, cardiopulmonary reserve): Equal-weight average of VO_2peak %pred_v2 (improved reference equation), O_2 pulse peak (%pred), and peak METs; inverted to a risk score (higher = greater risk).

T-axis (Threshold, ventilatory threshold stability): Percentile scores for VT1/VO_2peak ratio and V_E/V_{CO_2} slope, capturing aerobic metabolic stability and ventilatory efficiency.

I-axis (Instability, haemodynamic instability): Percentile scores for peak systolic blood pressure and O_2 pulse trajectory (peak-slope proxy), augmented by an EIH status penalty term.

2.5.2 Composite Score and Threshold Determination

The three axes are combined with equal weighting (≈1/3 each) into a continuous S_{lab} score (0–100, higher = greater risk):

$$S_{\text{lab}} = \frac{1}{3}(s_R + s_T + s_I) \tag{2}$$

Equal weighting ensures score computation is independent of any external clinical outcome, eliminating circular dependency. Thresholds are set by reference population percentiles: the 75th (P75) and 90th (P90) percentiles of S_{lab} in the reference subset (CTRL, n=962) define the Green/Yellow and Yellow/Red boundaries, respectively.

2.5.3 Threshold Robustness Validation

Three complementary approaches validated threshold robustness. **Bootstrap confidence intervals:** 1000 bootstrap resamples (with replacement) estimated 95% CIs for both thresholds; CI width <2 indicates high stability. **Sensitivity scanning:** Both thresholds were offset by ±5% and ±10%, and zone redistribution was recorded (reclassification rate <10% considered robust). **Literature threshold comparison:** Data-driven VO_2peak%pred at each threshold was compared against Weber-Janicki criteria [8] (Green ≥70%, Yellow 50–70%, Red <50%); deviation >10% indicates necessity of population-specific calibration. A reclassification matrix against the fixed-threshold zone was computed, with emphasis on Red↔Green crossover cases.

2.6 Statistical Analysis

Baseline characteristics were compared with Kruskal-Wallis tests (continuous variables; Shapiro-Wilk for distribution assessment) and χ^2 tests (categorical variables), with Dunn's post-hoc pairwise comparisons (Bonferroni correction, α=0.0083). HTN×EIH interactions were assessed by Type II ANOVA (statsmodels); effect sizes were quantified as partial η^2 [14] (small: 0.01–0.06; medium: 0.06–0.14; large: >0.14). Given N=3,206, effect size rather than p-value was the primary interpretive metric. As auxiliary validation, machine learning models (LASSO/XGBoost/LightGBM/CatBoost) were used to evaluate the predictive performance ceiling of summary-level CPET data, with a data leakage prevention mechanism to ensure valid benchmarking; see Supplementary Materials for details.

3 Results

3.1 Baseline Characteristics

Median age was 67 years across all four cohorts, with no significant between-group difference (p=0.375), indicating balanced age distributions (Table 2). HTN cohorts (HTN-NoEIH and HTN-EIH) had significantly higher body weight than non-HTN groups (p<0.001), consistent with the metabolic syndrome background of hypertension. HTN-NoEIH showed the lowest median VO_2peak (19.2 mL/kg/min), EIH-Only the highest (23.5 mL/kg/min), and HTN-EIH had the highest peak systolic blood pressure (214 mmHg). Post-hoc comparisons identified significant between-group differences in 10/11 key variables, suggesting that HTN and EIH affect cardiopulmonary function through distinct mechanisms.

Table 2: Baseline characteristics of the four cohorts

Variable	CTRL (n=1,858)	HTN-NoEIH (n=738)	HTN-EIH (n=277)	EIH-Only (n=333)	P-value[a]
Continuous variables [median (Q1, Q3)]					
Age (years)	67 [63,72]	67 [63,71]	67 [63,71]	67 [64,71]	0.375
VO_2peak (mL/kg/min)	23.2 [19.5,28.3]	19.2 [16.2,23.0]	20.1 [16.8,23.4]	23.5 [19.3,28.9]	<0.001
VO_2peak%pred (%)	77.0 [66,88]	72.0 [62,86]	80.0 [67,91]	79.0 [68.5,92]	<0.001
Peak HR (bpm)	154 [140,167]	134 [119,148]	145 [129,158]	158 [145,170]	<0.001
Peak SBP (mmHg)	161 [144,178]	171 [155,184]	214 [204,226]	212 [200,223]	<0.001
V_E/V_{CO_2} slope	26.7 [24.2,29.4]	27.8 [25.3,30.8]	27.6 [24.8,30.0]	27.4 [24.4,30.0]	<0.001
O_2 pulse peak (mL/beat)	10.0 [8.0,13.0]	10.0 [8.2,13.0]	10.0 [8.0,13.0]	11.0 [8.0,14.0]	0.036
Categorical variables (n, %)					
Female sex	926 (49.8%)	312 (42.3%)	135 (48.9%)	147 (44.1%)	0.004
Hypertension history	0 (0%)	738 (100%)	277 (100%)	0 (0%)	<0.001
EIH	0 (0%)	0 (0%)	277 (100%)	333 (100%)	<0.001

[a] Kruskal-Wallis (continuous) or χ^2 (categorical); Dunn's post-hoc with Bonferroni correction. Comorbidity fields (diabetes, coronary artery disease) were uniformly zero in the current dataset (data limitation).

3.2 HTN×EIH Interaction Analysis

The baseline differences suggested that HTN and EIH influence cardiopulmonary function through distinct mechanisms; Type II two-way ANOVA further quantified the main effects and interaction (Table 3).

HTN history was the primary driver of reduced VO_2peak (η^2=0.084, medium effect) and peak HR (η^2=0.133, medium effect), reflecting systemic suppression of exercise capacity and chronotropic function by hypertension. EIH manifested primarily as a markedly elevated peak SBP (mean 214.85 vs 160.76 mmHg, η^2=0.371, large effect), with minimal

Table 3: Two-way ANOVA for HTN×EIH effects

Outcome	HTN main effect			EIH main effect			Interaction (A×B)	
	F	p	η^2	F	p	η^2	F (p)	η^2
VO_2peak	290.6	<0.001	0.084^m	4.37	0.037	0.001^n	0.02 (0.894)	0.000^n
VO_2peak%pred	22.3	<0.001	0.007^n	25.9	<0.001	0.008^n	5.03 (0.025)	0.002^n
Peak HR	487.6	<0.001	0.133^m	42.4	<0.001	0.013^s	12.87 (<0.001)	0.004^n
Peak SBP	65.6	<0.001	0.021^s	1803.1	<0.001	0.371^L	5.20 (0.023)	0.002^n
V_E/V_{CO_2} slope	38.2	<0.001	0.012^s	0.34	0.558	0.000^n	3.37 (0.067)	0.001^n
O_2 pulse peak	2.91	0.088	0.001^n	9.67	0.002	0.003^n	5.84 (0.016)	0.002^n

Effect size: ^{n} negligible (<0.01), ^{s} small (0.01–0.06), ^{m} medium (0.06–0.14), ^{L} large (>0.14) (Cohen 1988 [14]).

independent impact on VO_2peak (η^2=0.001). The HTN×EIH interaction for VO_2peak was non-significant (F=0.02, p=0.894, η^2=0.000); all statistically significant interactions in this large sample had negligible effect sizes. These findings indicate that HTN and EIH operate through the cardiopulmonary reserve axis (R-axis) and the haemodynamic instability axis (I-axis) independently, providing the physiological basis for separate multi-axis evaluation.

3.3 Improved Reference Equations

Given that HTN exerts an independent main effect on cardiopulmonary reserve, accurate R-axis assessment requires an improved reference equation. Using the reference subset (CTRL, n=962; male=480, female=489), we constructed the improved reference equations (Table 4).

Table 4: Reference equation v1 vs v2 comparison

Target variable	v1 R^2	v2 R^2 (fit)	v2 R^2 (5-fold CV)	ΔR^2	N
VO_2peak (mL/kg/min)	0.298	**0.392**	0.378	+0.094	962
Peak HR (bpm)	0.050	0.101	0.091	+0.051	960
V_E/V_{CO_2} slope	0.030	0.061	0.033	+0.031	963
O_2 pulse peak (mL/beat)	0.200	0.511	0.506	+0.311	962

v2 adds BMI to the v1 formula (age + sex + interaction term).
VO_2peak improved equation (Eq. 1):

$$\widehat{VO_2\text{peak}} = 41.10 - 0.058 \times \text{age} + (-1.877 + 0.163 \times \text{age}) \times \mathbf{1}_M - 0.606 \times \text{BMI}.$$

BMI incorporation raised VO_2peak R^2 from 0.298 to 0.392 (5-fold CV=0.378, ΔR^2=+0.094), with the most pronounced improvement for O_2 pulse (R^2: 0.200 to 0.511). External equation validation (Table 5) showed that both Wasserman & Hansen (1999) and Koch et al. (2009) yielded $R^2 < 0$ in this population, with systematic bias reaching −11.85

mL/kg/min, confirming that European-derived reference standards are not applicable to Chinese elderly hypertensive patients and that a locally derived reference subset is necessary.

Table 5: External reference equations applied to the current dataset (VO_2peak)

External equation	R^2	RMSE	Systematic bias	N
Wasserman & Hansen 1999 (male) [10]	−0.650	9.50	−5.55	479
Wasserman & Hansen 1999 (female) [10]	−5.435	12.89	−11.85	489
Koch et al. 2009 (male) [12]	−0.162	7.97	+1.86	479
Koch et al. 2009 (female) [12]	−0.016	5.12	−0.00	489

$R^2 < 0$ indicates that the reference equation performs worse than a mean-only predictor. Negative bias = equation overestimates measured values.

3.4 Data-Driven Safety Zone Distribution, Threshold Validation, and Literature Comparison

Following construction of the R/T/I three-axis scores from the improved reference equation, the full-sample safety zone distribution (Table 6) showed: Green 1,316 (41.0%), Yellow 544 (17.0%), Red 1,346 (42.0%). HTN-EIH had the highest Red proportion (89.5%), consistent with its high I-axis haemodynamic stress; CTRL had the most favourable distribution (Green 51.8%, Red 29.4%).

Table 6: Data-driven safety zone distribution

Cohort	N	Green (%)	Yellow (%)	Red (%)
CTRL	1,858	962 (51.8)	350 (18.8)	546 (29.4)
HTN-NoEIH	738	325 (44.0)	147 (19.9)	266 (36.0)
HTN-EIH	277	11 (4.0)	18 (6.5)	248 (89.5)
EIH-Only	333	18 (5.4)	29 (8.7)	286 (85.9)
Male	1,678	741 (44.2)	295 (17.6)	642 (38.3)
Female	1,520	575 (37.8)	248 (16.3)	697 (45.9)
Total (N=3,206)	3,206	1,316 (41.0)	544 (17.0)	1,346 (42.0)

Global thresholds: Green/Yellow boundary = 24.17 (S_{lab} points), Yellow/Red boundary = 27.13. Reference percentile method (P75/P90); reference subset n=962 (CTRL).

Bootstrap confidence intervals (Table ??A) confirmed highly stable thresholds: Green/Yellow CI width 0.65, Yellow/Red CI width 0.94, both <1.0. Sensitivity scanning further confirmed robustness, with reclassification rates of 7.3–7.5% under ±5% offsets, well below the 10% criterion. Systematic comparison with Weber-Janicki criteria (Table ??B) revealed an important asymmetry: the Yellow/Red boundary corresponded to VO_2peak%pred=71%,

a +21% deviation from the published 50% threshold; the Green/Yellow boundary showed only a +5% deviation, indicating reasonable cross-population agreement for the low-risk identification boundary. This asymmetric pattern implies that directly applying published criteria would overclassify approximately 21% of patients with VO_2peak%pred 50–70% as Yellow risk when they in fact belong to the Green zone in this population.

Table 7: Threshold robustness validation and literature comparison

A. Bootstrap 95% confidence intervals (N=1000 resamples)

Threshold	Reference value	95% CI	CI width	Stability
Green/Yellow (P75)	24.17	[23.87, 24.53]	0.65	✓Highly stable (<1)
Yellow/Red (P90)	27.13	[26.64, 27.58]	0.94	✓Highly stable (<1)

B. Data-driven thresholds vs Weber-Janicki literature criteria

Metric	Literature (Weber-Janicki)	Data-driven	Deviation	Interpretation
VO_2peak%pred @ Green/Yellow	70.0%	75.0%	+5.0%	✓Largely consiste
VO_2peak%pred @ Yellow/Red	50.0%	71.0%	+21.0%	× Substantial devi
V_E/V_{CO_2} slope @ Yellow/Red	36.0	27.3	−8.7	× Substantial devi

Literature thresholds from Weber & Janicki (1986) [8]. CI width <1.0 indicates extreme threshold robustness. Yellow/Red +21% deviation indicates systematic bias in this elderly hypertensive population, necessitating population-specific calibration.

The reclassification matrix (Table 9) quantified the clinical impact of threshold migration: compared with fixed-threshold zones, the overall agreement rate was 60.5% with a reclassification rate of 39.5% (1,258/3,185), including 188 Red↔Green crossover cases (5.9%) that require priority clinical review.

Table 8: Reclassification matrix: data-driven zone vs fixed-threshold zone

Old zone ↓ New zone →	Green	Yellow	Red	Subtotal
Green	958 (30.1%)	186 (5.8%)	145 (4.6%)†	1,289
Yellow	311 (9.8%)	285 (8.9%)	503 (15.8%)	1,099
Red	43 (1.4%)†	69 (2.2%)	685 (21.5%)	797
Total	1,312	540	1,333	3,185

Agreement rate: 60.5%; reclassification rate: 39.5% (1,258/3,185). †: Crossover reclassification (Red↔Green), 145+43=188 cases (5.9%), requiring clinical review.

3.5 Auxiliary Validation

As methodological auxiliary validation, machine learning models (pre-exercise risk prediction AUC=0.583; post-exercise safety classification F1=0.499) performed comparably

to a 2-variable clinical baseline (HTN history + BMI, AUC=0.611), failing to exceed traditional clinical features—reflecting the predictive performance ceiling of summary-level CPET data alone. Proxy leakage ablation analysis confirmed negligible leakage contribution from VO_2peak (ΔF1=+0.003), indicating that the performance ceiling reflects intrinsic feature set limitations rather than methodological flaws. Multi-definition concordance analysis (K=4 definitions) identified 26.2% high-confidence classifications (841 patients); 73.5% showed cross-definition discordance, reflecting structural uncertainty at the summary data level. High-confidence Red zone event rate (28.6%) was substantially higher than Green zone (5.1%), validating reliable clinical discrimination in high-confidence classifications. Detailed model results and supplementary methods are provided in Supplementary Materials.

4 Discussion

4.1 Core Findings and Clinical Implications

This study systematically established a data-driven R/T/I three-axis exercise safety zone framework in a cohort of 3,206 elderly hypertensive patients. The core methodological contribution lies in integrating multidimensional cardiopulmonary information—reserve (R-axis), ventilatory threshold stability (T-axis), and haemodynamic instability (I-axis)—into a continuous composite score anchored to reference population distribution boundaries. The fundamental advantage of equal weighting is that threshold computation is entirely independent of any external clinical outcome, avoiding circular dependency; robustness is formally quantified by Bootstrap CI (Green/Yellow 0.65, Yellow/Red 0.94, both <1.0)—a metric rarely reported in existing CPET safety zone literature. This framework extends beyond the limitation of single-metric (VO_2peak%pred) safety zone definitions [8], providing a reproducible paradigm for systematic integration of multidimensional CPET information.

The +21% systematic deviation of the published Yellow/Red threshold is the most clinically significant finding of this study. This deviation implies that directly applying Weber-Janicki criteria to Chinese elderly hypertensive patients would overestimate risk for approximately 21% of patients with VO_2peak%pred 50–70%, misclassifying them as Yellow when they belong to Green in this population. The reclassification rate of 39.5% and 188 Red↔Green crossover cases quantify the clinical consequences of this threshold mismatch. The contrasting finding—that the Green/Yellow boundary shows only a +5% deviation—suggests that the low-risk identification boundary transfers reasonably well across populations, while the moderate-to-high risk boundary is the critical calibration target. This asymmetric deviation pattern closely aligns with the call for localised

threshold validation in the Chinese expert consensus [9], providing empirical evidence for multi-centre prospective calibration studies.

The improved reference equation incorporating BMI (ΔR^2=+0.094) is theoretically well supported: BMI serves as a readily available body composition proxy, and both reduced muscle mass and increased fat fraction—prevalent in elderly hypertensive patients—are independently associated with diminished cardiopulmonary reserve. The BMI coefficient of −0.606 mL/kg/min per unit is consistent with published associations between obesity and reduced exercise capacity in older adults. The failure of external equations ($R^2 < 0$, maximum systematic error −11.85 mL/kg/min) further confirms that even equations derived from cohorts including Asian sub-samples (Koch et al. 2009) cannot be directly transferred to Chinese elderly hypertensive patients, establishing the necessity of a local reference subset (CTRL, n=962) as the foundation for reliable R-axis assessment.

The finding that HTN and EIH exert independent main effects on distinct cardiopulmonary dimensions provides the physiological rationale for multi-axis framework design. HTN's medium-effect impacts on exercise capacity (η^2=0.084) and chronotropic function (η^2=0.133) reflect systematic suppression of cardiac output reserve and autonomic regulation by hypertension [4]; EIH's large-effect elevation of peak SBP (η^2=0.371) reveals a distinct haemodynamic stress mechanism characteristic of EIH patients. The non-significant HTN×EIH interaction for VO_2peak (p=0.894) indicates that the two factors exert independent additive rather than synergistic effects on cardiopulmonary reserve, statistically supporting the principle of evaluating R and I axes separately and independently.

4.2 Comparison with Prior Literature

At the methodological level, this study complements the CPET modelling data leakage prevention framework proposed by Ross et al. [13] and the breath-by-breath VT detection research of Zignoli et al. [15]: the latter achieves fine-grained stratification from raw waveform data, while this study establishes a robust safety zone threshold methodology from summary-level data, representing complementary CPET safety assessment approaches at different data granularities.

For reference equations, Koch et al. [12] SHIP-derived values (including a Chinese subsample) yielded $R^2 \approx -0.16$ (male) and −0.02 (female) in this population, confirming that even equations from partly Asian reference populations cannot be applied to Chinese elderly hypertensive patients and reinforcing the necessity of locally derived reference subsets. Compared with the Chinese expert consensus [9], this study provides the first large-sample empirical calibration of the recommended VO_2peak%pred thresholds in an elderly hypertensive cohort.

Machine learning performance (AUC≈0.58, F1≈0.50) was lower than that of Li et

al. [16] (AUC=0.794) and Arina et al. [17] (AUC=0.946). This performance gap reflects systematic feature set differences—our study lacked medication data, left ventricular ejection fraction, and other key clinical variables—rather than methodological deficiencies. This discrepancy in fact substantiates positioning the R/T/I safety zone framework (not an ML prediction tool) as the core contribution: the performance ceiling of summary-level CPET data alone is insufficient for clinical prediction tasks, yet precisely adequate for establishing stable, data-anchored stratification thresholds.

4.3 Limitations

The principal limitations of this study are as follows: (1) single-centre retrospective design with potential selection bias and absence of an independent external validation cohort; (2) complete absence of medication data—β-blocker use significantly attenuates HR responses and may confound HR-related metrics; (3) EIH classification derived from group-coding fields rather than continuous real-time SpO_2 monitoring, introducing potential misclassification; (4) exclusive use of summary-level CPET data without breath-by-breath waveforms, constraining accurate VT/RCP detection and the predictive performance ceiling; (5) equal weighting of R/T/I axes—the relative contribution of each axis requires prospective validation; (6) reference equation R^2=0.392 retains substantial residual variance, and the oldest subgroup ($\geq$75 years) warrants further validation.

4.4 Future Directions

Future research should incorporate breath-by-breath gas exchange and 12-lead ECG waveform data to overcome the predictive ceiling of summary-level data and to enable accurate VT1/RCP detection at the waveform level, providing more reliable T-axis inputs [15, 18]. Multi-centre prospective validation is the critical next step for generalising this framework and should incorporate complete medication data (particularly β-blockers) to correct related biases, and independently validate age- and sex-stratified thresholds. The ability of wearable devices to proxy key CPET anchor variables (VO_2peak, HR_{peak}, peak SBP) warrants systematic exploration [19], to evaluate the feasibility of home exercise safety monitoring.

5 Conclusions

This study establishes the first data-driven R/T/I three-axis exercise safety zone framework, with reference population percentile-based thresholds validated by Bootstrap as highly robust (CI width <1.0 for both boundaries), providing a reproducible data-driven

paradigm for CPET safety zone quantification in Chinese elderly hypertensive patients. The published Yellow/Red threshold shows a +21% systematic deviation in this population (Green/Yellow only +5%), necessitating population-specific calibration for moderate-to-high risk stratification; a reclassification rate of 39.5% quantifies the clinical cost of direct application of literature-derived thresholds. The BMI-enhanced improved reference equation substantially outperforms traditional equations (ΔR^2=+0.094), and existing external reference equations ($R^2 < 0$) cannot be applied directly to this population. Future studies incorporating breath-by-breath waveform data and multi-centre prospective validation are needed to further enhance the clinical utility and generalisability of this framework.

Figure Legends

Figure 1: HTN×EIH two-way interaction plot for VO_2peak (group means). CTRL vs HTN group difference is significant (η^2=0.084, medium effect); EIH+/EIH− groups show comparable VO_2peak within the same HTN status, with a non-significant interaction (p=0.894).

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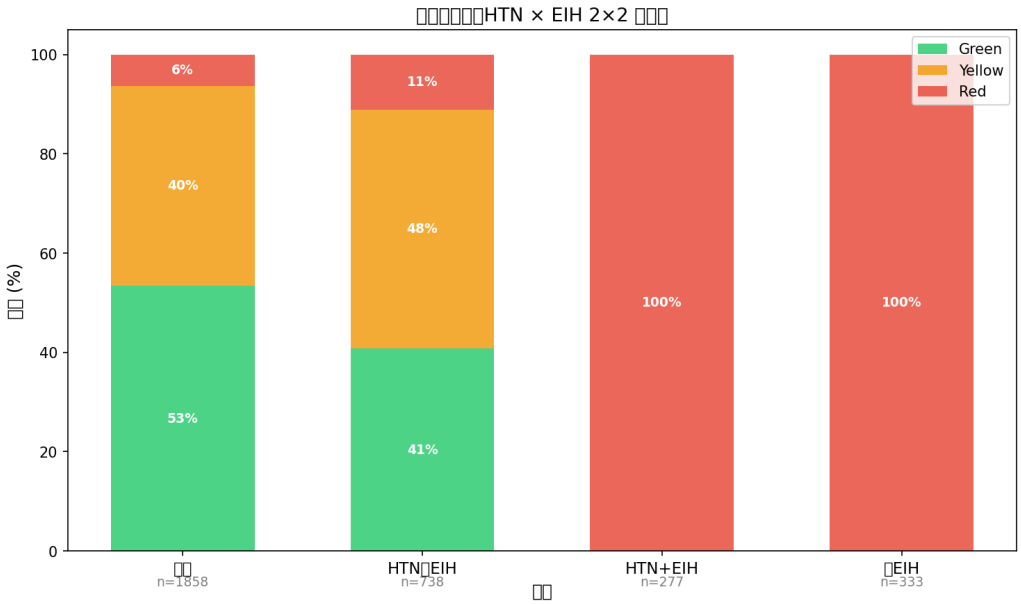

Figure 2: Data-driven safety zone distribution (Green/Yellow/Red) as stacked bar charts across the HTN×EIH 2×2 cohorts. HTN-EIH and EIH-Only cohorts show markedly higher Red proportions (89.5%/85.9% vs 29.4% in CTRL), reflecting the dominant I-axis haemodynamic instability effect of EIH.

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